Presentation-aware E-values, cross-allele pseudosequence diffusion, and proteome-scale peptide matching for peptide–MHC analysis

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mhcmatch technical appendix

Abstract

This appendix develops the mathematical and statistical theory of `mhcmatch`, the applied peptide–MHC layer built on the `seqtree` fuzzy-search substrate. We take as given the control-calibrated E-value of the `seqtree` appendix (`evalue.tex`) and specialize it to MHC *presentation*, whose null is allele-conditional and anchor-constrained. We restate the anchor / TCR-facing decomposition and the per-allele presentation-aware E-value (the *forward* restriction problem), then the *reverse* problem—peptide to presenting allele—as a neighbour-vote posterior with a binomial-tail confidence E-value, together with its non-binder filter and class-II promiscuity treatment. The central contribution is a *cross-allele diffusion* model: each allele is represented by its 34-residue groove pseudosequence, and per-anchor preference distributions are smoothed by kernel-weighted empirical-Bayes shrinkage toward groove-similar alleles, with the per-anchor relevance of each groove position *learned* from data (a mutual-information feature importance that separates, e.g., the MHC-I B-pocket governing P2 from the F-pocket governing PΩ). This rescues rare alleles with few peptides—lifting the `seqtree` limitation that distinct alleles are distinct, never-pooled nulls—and we give its hierarchical-Bayes interpretation, effective sample size, limiting cases, and bias–variance trade-off. We then treat proteome-scale scanning (sliding-window presentation with multiple-testing control; near-exact neoantigen source identification), motif logos with length distributions, the composition rules for the planned stability / affinity / cleavage / expression / immunogenicity predictors, and the benchmark protocol against NetMHCpan, NetMHCIIpan, MixMHCpred and MixMHC2pred.

1. Scope and relation to the substrate

`seqtree` provides a payload-agnostic fuzzy-search core and a control-calibrated E-value: for a query q and a scope θ, with an empirical background P_0 estimated from a matched control of size M and

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a target set of size N, the expected number of chance neighbours is $\widehat{E}(q,\theta) = (N/M)\, n_C(q,\theta)$, with $p_{\text{any}} = 1 - e^{-\widehat{E}}$ and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\widehat{E}) \geq n_D)$ (`seqtree` appendix, Def. of the E-value; Poisson/Chen–Stein bound; Karlin–Altschul [9, 1] as the i.i.d. special case). We do not re-derive that theory; we *specialize* it.

For peptide–MHC the relevant background is not V(D)J generation but *presentation*, which is allele-conditional and constrains the anchor positions. `mhcmatch` owns four developments on top of the substrate: (i) the productionized forward/reverse presentation E-value (§ 2–3); (ii) the cross-allele *pseudosequence diffusion* model that pools statistics across groove-similar alleles (§ 4, the headline); (iii) proteome-scale scanning and near-exact source identification (§ 5); and (iv) motif logos (§ 6). Planned predictors and the comparison protocol are § 7–8; limitations are § 9.

2. Anchor decomposition and the per-allele presentation E-value

A presented peptide of length L factorizes $\sigma = (\sigma_A, \sigma_T)$ into MHC-facing *anchors* $A(a, L)$ and TCR-facing positions T. For class I the buried anchors are $\{\mathrm{P2}, \mathrm{P\Omega}\}$ (the B- and F-pockets); for the class II 9-mer core they are $\{\mathrm{P1}, \mathrm{P4}, \mathrm{P6}, \mathrm{P9}\}$. *Presentation* fixes σ_A; *recognition* reads σ_T. Dolton et al. [7] exhibit one HLA-A02:01 TCR cross-reacting with three epitopes through a shared central motif with the anchors irrelevant, which motivates two complementary maskings, each realized by a per-position weight vector w over a `seqtree` `PositionalMatrix`:

$$\text{(recognition)} \quad w_i = \mathbb{1}[i \in T]; \qquad \text{(presentation)} \quad w_i = \mathbb{1}[i \in A(a, L)]. \tag{1}$$

In `mhcmatch` these are exposed by `Store.decompose`, which returns both readouts with the masked positions written as the spare symbol `X`.

Forward problem.. For a candidate restriction allele a with an anchor-masked presented control C_a of size M_a and target peptidome of size N_a, the presentation-aware E-value is the substrate E-value on allele-restricted, anchor-conditioned indices,

$$\widehat{E}(q, \theta; a) = \frac{N_a}{M_a}\, n_{C_a}(\sigma_T, \theta), \qquad p_{\text{enrich}} = \mathbb{P}(\,\text{Poisson}(\widehat{E}) \geq n^T_{D_a}), \tag{2}$$

i.e. significance is computed on the TCR-facing readout only, so shared anchors (presentation, not recognition) do not inflate within-allele hits. The analytic null for rare a is taken up in § 4.

3. Reverse problem: peptide to presenting allele

The forward E-value answers "is q a binder of a *given* a?". Restriction prediction asks the reverse: which alleles present q? We flip the weights in (1) to the *presentation* mode and index reference peptides by their anchor signature (`layout.presentation_features`; class I N/C-pocket residues P1 P2 P3, P(Ω—1), PΩ, class II core P1 P4 P6 P9). For a query we widen the scope on its signature until it has $n \in [10, 100]$ non-exact neighbours, and tally their alleles, k_a votes out of n.

Definition 1 (Ranking and confidence). The *ranking* score is the neighbour vote fraction $\hat{p}(a \mid q) = k_a/n$, a posterior over the panel. The *confidence* score is the per-allele enrichment over the panel background frequency f_a,

$$E_a = -\log_{10} \mathbb{P}(\text{ Binomial}(n, f_a) \geq k_a), \tag{3}$$

the upper binomial tail; a peptide binds a iff $E_a \geq -\log_{10} \alpha$ and $k_a > 0$.

Remark 1 (Why vote fraction, not enrichment, ranks). On a skewed panel the dominant true allele has a large expected count nf_a, so the enrichment (3) *penalises* it; the vote fraction does not. Hence $\hat{p}$ ranks and E_a gates. This separation, validated on `pmhc_data` (per-(peptide, allele) ROC-AUC 0.90–0.99, `seqtree/bench/bench_mhc_guess.py`), is what `Store.restriction` implements.

Class-II register trick.. The class-II core register inside a longer peptide is unknown; full deconvolution clusters it [3, 2]. We use a one-pass proxy: pick the single 9-mer window whose P1 is a large hydrophobic and whose P4/P6/P9 avoid Pro/Gly, and take that core's {P1,P4,P6,P9}. Committing to one register makes an allele's peptides share a consistent signature and is what lifts class-II ROC-AUC into the 0.9 range.

Non-binder filter and promiscuity.. A peptide that binds *no* panel allele has small $\max_a E_a$ (real-vs-random rejection); one that does not bind a *specific* a has $E_a < -\log_{10} \alpha$. The global statistic $E_{\text{glob}} = \sum_a 10^{-E_a}$ is the expected number of panel alleles presenting q by chance. Class II is genuinely multi-label (an open groove with loosely specified pockets presents one peptide on many alleles), so the panel positive set is *every* observed restricting allele and the non-binder test is "binds none of the panel".

4. MHC pseudosequence model and cross-allele diffusion

`seqtree` deliberately treats distinct alleles as distinct, never-pooled nulls. This is correct but wasteful: rare alleles carry few presented peptides, so both (2) and (3) are noisy exactly where data are thin—the equity problem of Glynn et al. [8]. `mhcmatch` lifts the limitation by making the allele a *sequence* the same engine can compare, and diffusing statistics along groove similarity.

Pseudosequence.. Each allele a is represented by its NetMHCpan-style 34-residue groove pseudosequence $s_a \in \Sigma^{34}$ [13]: the polymorphic, peptide-contacting positions (class I α_1/α_2; class II $\alpha_1 + \beta_1$). Per-allele binding can be predicted from s_a alone [8, 4], so groove distance is a meaningful allele distance.

4.1. Anchor-factored similarity kernel

The presentation motif factorizes by pocket: anchor j's residue preference is governed by the groove positions lining pocket j, and different pockets are spatially separated (the class-I B-pocket sets P2, the F-pocket sets PΩ). We therefore use a *per-anchor* kernel rather than one global allele distance. For anchor j let $w_j \in \mathbb{R}^{34}_{\geq 0}$ weight the groove positions, and define

$$d_j(a, b) = \sum_{p=1}^{34} w_{j,p}\, \delta(s_{a,p}, s_{b,p}), \qquad K_j(a, b) = \exp(-d_j(a, b)/h_j), \tag{4}$$

where $\delta(x, y) = s(x, x) + s(y, y) - 2s(x, y)$ is the BLOSUM62 Gram distance (zero on identity, small for conservative substitutions, large for dissimilar residues; normalized so an average substitution costs ≈ 1), evaluated through seqtree's substitution matrix (`SubstitutionMatrix.penalty`, exposed for this use). Bandwidth $h_j > 0$; ambiguous positions X are skipped. Replacing δ with the identity indicator $\mathbb{1}[x \neq y]$ gives the plain Hamming variant, and $w_j \equiv 1$ a single un-factored kernel.

Definition 2 (Learned per-anchor relevance). Let A_j be the allele's modal residue at peptide anchor j (from its presented set) and S_p the groove residue at pseudosequence position p, both viewed as categorical variables over the allele panel. The relevance of position p to anchor j is the mutual information

$$w_{j,p} \;\propto\; \mathrm{I}(S_p;A_j) = \sum_{x,y} \hat{\mathbb{P}}(S_p{=}x, A_j{=}y)\;\log_2 \frac{\hat{\mathbb{P}}(S_p{=}x, A_j{=}y)}{\hat{\mathbb{P}}(S_p{=}x)\,\hat{\mathbb{P}}(A_j{=}y)}, \tag{5}$$

normalized to mean one over p. Equivalently w_j may be fit by maximizing the held-out predictive log-likelihood of the shrinkage model below under an ℓ_1/entropy penalty, which selects each anchor's pocket positions. (5) is the "feature importance" that says which groove residues drive P2 versus PΩ.

4.2. Kernel-shrinkage pooling

Write $\theta_{a,j}$ for allele a's empirical residue distribution at anchor j over its n_a peptides. We shrink it toward groove-similar alleles:

$$\hat{\theta}_{a,j} = \frac{n_a\,\theta_{a,j} + \sum_{b\neq a} K_j(a,b)\,n_b\,\theta_{b,j}}{n_a + \sum_{b\neq a} K_j(a,b)\,n_b}, \qquad \mathrm{ESS}_{a,j} = n_a + \sum_{b\neq a} K_j(a,b)\,n_b. \tag{6}$$

Proposition 1 (Hierarchical-Bayes reading and limits). *(6) is the posterior mean of a Dirichlet–multinomial in which the neighbour-weighted counts form the prior pseudocounts: $\hat{\theta}_{a,j} = \mathbb{E}[\theta \mid counts]$ with concentration $\mathrm{ESS}_{a,j}$. As $h_j \to 0$, $K_j(a,b) \to 0$ for $b \neq a$ and $\hat{\theta}_{a,j} \to \theta_{a,j}$ (no pooling); as $h_j \to \infty$, $K_j(a,b) \to 1$ and $\hat{\theta}_{a,j} \to (\sum_b n_b\theta_{b,j})/\sum_b n_b$ (the global pool).*

(Both limits are implemented and unit-tested in `Pseudoseq.shrink`.)

Proposition 2 (Bias–variance and bandwidth). *Treat each $\theta_{b,j}$ as an unbiased estimate of allele b's truth $\theta^\star_{b,j}$ with variance $\propto 1/n_b$. The pooled estimator has variance $O(1/\mathrm{ESS}_{a,j})$ and bias $\|\sum_b \kappa_b(\theta^\star_{b,j} - \theta^\star_{a,j})\|$, where $\kappa_b = K_j(a,b)n_b/\mathrm{ESS}_{a,j}$ is the borrowed mass. The mean-squared error is minimized by borrowing more (smaller effective h_j penalty) when n_a is small and groove neighbours are close, and less when n_a is large; calibrating h_j by cross-validated predictive likelihood realizes this trade-off automatically.*

Bounded-concentration variant (the practical default).. The counts-weighted form (6) lets one deep neighbour ($K_j n_b \gg n_a$) overwrite a rare allele's own peptides. We therefore default to a fixed prior strength τ: shrink toward the kernel-weighted neighbour mean $m_{a,j} = (\sum_b K_j(a,b)n_b\theta_{b,j})/\sum_b K_j(a,b)n_b$,

$$\hat{\theta}_{a,j} = \frac{n_a\,\theta_{a,j} + \tau\,m_{a,j}}{n_a + \tau}, \tag{7}$$

so borrowing adds at most τ pseudocounts and vanishes automatically as n_a grows (Prop. 2). Empirically (`bench/bench_diffusion.py`; human, allele-split held-out rank AUC with the BLOSUM-scored, MI-weighted kernel (4), $\tau = 10$, $h = 2$, the per-allele score being the anchor log-odds over the N/C pocket footprint {P1,P2,P3,P(Ω—1),PΩ}) the diffusion lifts the AUC of *rare* MHC-I alleles (≤ 30 peptides) from 0.82 to 0.94 on the high-confidence (≥ 2-publication) set while leaving *frequent* alleles unchanged ($0.975 \to 0.975$); on the noisier full set the rare-allele gain is $+0.06$ and frequent alleles are never harmed (Fig. 1). Stats are computed fairly: held-out peptides are excluded from *every* allele's training set, so an identical co-presented copy cannot leak into a rare allele's score through diffusion (this lowers the raw baseline and *widens* the honest gain). The richer pocket footprint matters: with the two primary anchors alone ({P2,PΩ}) the full-set rare gain nearly vanishes, whereas the auxiliary pocket-proximal positions make per-allele estimation sparser—so diffusion has more to rescue. The data-rescue is safe: it never lowers a well-sampled allele's AUC.

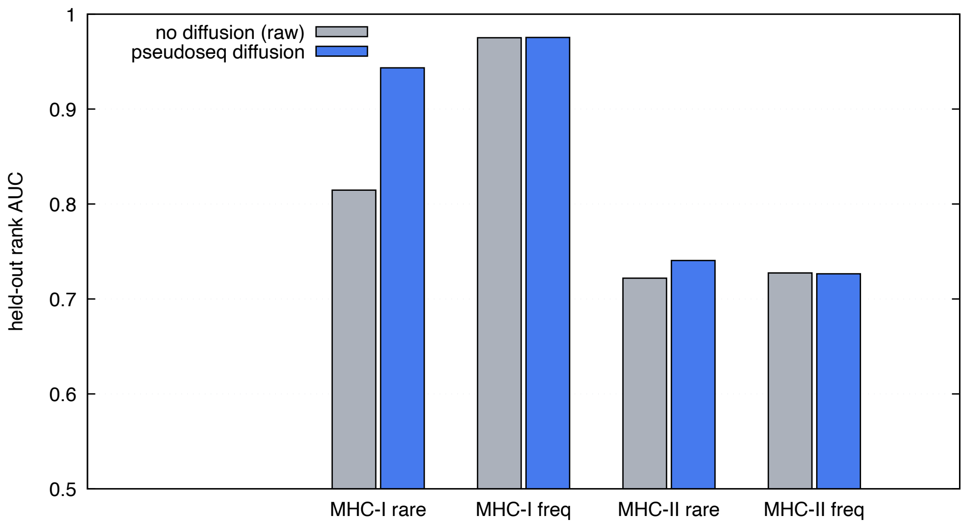

Figure 1: Held-out rank AUC for rare (≤ 30 peptides) and frequent (≥ 200) alleles, without (raw, grey) and with (blue) pseudosequence diffusion, allele-split. MHC-I on the ≥ 2-publication set, MHC-II on the full set. Diffusion lifts rare alleles and leaves frequent alleles unchanged in both classes. `bench/bench_diffusion.py`.

Bayesian reading (in plain terms).. The shrinkage (7) is exactly a Bayesian posterior mean. Treat each allele's anchor-j preference $\theta_{a,j}$—the probabilities of each amino acid in pocket j—as *unknown*. Before seeing allele a's own peptides we expect it to resemble its groove neighbours, so we put a prior centred on the kernel-weighted neighbour mean $m_{a,j}$ with strength τ (measured in pseudo-peptides); the allele's n_a observed peptides then update that belief. With the conjugate Dirichlet–multinomial model

$$\theta_{a,j} \sim \mathrm{Dirichlet}(\tau\, m_{a,j}), \quad \sigma_{i,j} \mid \theta_{a,j} \sim \mathrm{Categorical}(\theta_{a,j}), \quad \theta_{a,j} \mid \{\sigma_{i,j}\} \sim \mathrm{Dirichlet}(\tau\, m_{a,j} + c_{a,j}), \tag{8}$$

where $c_{a,j}(r)$ counts peptides of a with residue r at pocket j ($\sum_r c_{a,j}(r) = n_a$), the posterior mean is *precisely* (7). A data-rich allele ($n_a \gg \tau$) ignores the prior and keeps its own motif; a data-poor one ($n_a \ll \tau$) leans on its neighbours. The bandwidth h (how far similarity reaches) and the strength τ (how much to borrow) are the only knobs (Fig. 2).

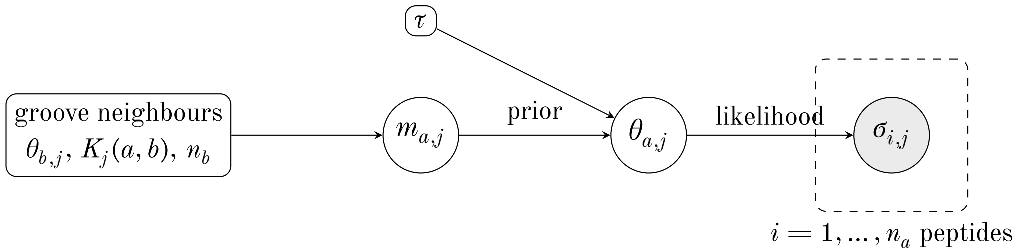

Figure 2: Graphical model behind the diffusion shrinkage. The kernel-weighted neighbour mean $m_{a,j}$ and the strength τ form a Dirichlet *prior* on the allele's pocket-j residue distribution $\theta_{a,j}$; the n_a observed anchor residues $\sigma_{i,j}$ (shaded) are the *likelihood*. The posterior mean is the shrinkage estimator (7): rare alleles borrow from groove-similar neighbours, well-sampled alleles do not.

Pooled null and restored E-value.. For a rare allele the analytic presented null is built from the shrunk anchor marginals: assuming approximate position independence in the anchor block (the maximum-entropy presented distribution of [14] conditioned on the motif), $\widehat{P_0}(\sigma_A \mid a) \approx \prod_j \hat{\theta}_{a,j}(\sigma_{A_j})$, which feeds a usable $\widehat{E}$ in (2)–(3) where the raw per-allele control was too small. The estimator is a smoothed null: its bias is controlled by the borrowed mass on dissimilar alleles (Prop. 2), vanishing as the panel around a becomes dense and groove-tight. In the production reverse problem (`Store.restriction`, `diffuse=True`) the shrunk anchor log-odds is used as the binder *gate*—it scores every requested allele, surfacing rare alleles with no signature neighbours—while the vote fraction (Def. 1) still *ranks*, since per-peptide anchor likelihood is diluted for broad, well-sampled alleles and is not itself a calibrated cross-allele ranking statistic.

4.3. Clustering and promiscuity

Thresholding K_j (or the un-factored K) yields single-linkage allele *clusters* of functionally similar alleles (`Pseudoseq.cluster`). A peptide presented across a cluster is *promiscuous*; its promiscuity is the spread of its positive alleles over the clustering, and the cluster is the natural unit over which to pool nulls when per-allele data are thin (especially class II). This is the principled cross-allele similarity that the substrate omits. Fig. 4 renders this as an allele network—edges are the MI-weighted, BLOSUM-scored kernel (4) above a soft/hard threshold, communities are greedy-modularity clusters [6]—which recover the classical HLA supertypes (A2, A3, B7, B27, B44, ...), the structural basis of cross-allele promiscuity. Mouse panels are few and groove-divergent, so they form no such communities.

The kernel induces an *allele similarity network* (Fig. 4): nodes are alleles, edges join pairs with $K \geq$ a soft threshold (bold above a hard threshold), and greedy modularity [6] recovers communities. On human MHC-I these communities reproduce the classical HLA supertypes (the A2, A3, A24, B7, B27, B44 groups), giving a data-driven, sequence-only account of the cross-allele promiscuity that lets one peptide be presented across a supertype. Mouse panels are small and the laboratory H-2 alleles are mutually divergent, so they form few or no communities — consistent with the thin, scattered mouse panels noted above. Rendered with Graphviz; `bench/promiscuity_graph.py`.

4.4. Per-pocket decomposition (class II)

The anchor-factored weights (5) make the pocket structure explicit. For each MHC-II core pocket P1/P4/P6/P9 we learn an independent weight vector w_j over the 34-residue class-II pseudosequence; Fig. 5 shows that the relevance concentrates on the polymorphic β_1 segment (positions 16–34) and is near-zero on the largely monomorphic α_1 segment—most sharply for DR, whose α chain (DRA) is invariant—while each pocket reads a *partly distinct* subset of those positions. The kernel is thus not one global allele distance but a sum of pocket-specific distances, each restricted to the groove residues that line its pocket. With this class-II model (alleles keyed by the α–β pair; cores from the register trick), diffusion gives a small but safe rare-allele gain ($0.75 \rightarrow 0.76$ AUC, frequent unchanged, Fig. 1); class II stays harder than class I (open groove, promiscuity, one-pass register).

5. Proteome-scale matching

Presentation scan.. To find all presented peptides in a protein we slide every binding-length window (class I 8–11; class II ≥ 13 with the register-anchored core) and evaluate (3) per window and allele. With W windows and $|\mathcal{A}|$ panel alleles the test is multiple: under the null the expected number of windows called for allele a is $\approx W\alpha$, so we control the family either by Bonferroni / max-statistic thresholds (FWER) or by Benjamini–Hochberg on the p_{enrich} across the $W \times |\mathcal{A}|$ grid (FDR) [5]. `scan_protein` reports per-window binders; the grid-level correction is a calibration step.

Same-MHC versus TCR-facing search.. At proteome scale two notions of "similar peptide" are distinguished by which mask of (1) is indexed: *same-MHC* similarity indexes the presentation signature (peptides likely presented by the same allele), *TCR-facing* similarity indexes the anchor-masked central motif (similar T-cell recognition, the basis of cross-reactivity). Both run on the C++ `KmerIndex` seed-and-gather.

Molecular mimicry.. A self or foreign peptide p is a candidate cross-reactive mimic of a neoantigen ν iff (i) it is presented by a compatible allele, (ii) the TCR-facing distance $d_{\text{TCR}}(p, \nu) = \sum_i w_i \, \text{pen}(p_i, \nu_i) \leq \theta$ (anchors zeroed), and (iii) the hit is significant against the per-allele presented background, $\widehat{E} \leq \alpha$. Condition (iii) deflates motifs common in the allele's peptidome and elevates surprisingly shared ones. This connects to the neoantigen-quality program of Łuksza et al.: the quality $Q = R \times D$ [11, 10] multiplies a recognition / non-self term R (the cross-reactivity distance to known antigens) by a self-discrimination term D (differential MHC binding), and to the close-to-self "shell" picture of [14]; `mhcmatch` supplies a calibrated E-value for the R component.

Near-exact source identification.. Given a neoantigen we identify the self peptide it derives from—its parent protein and the mutated position—by full-sequence (unmasked) $\leq m$-mismatch search over all length-L windows of the reference proteome. Under an i.i.d. background of composition $\{q_c\}$ the expected number of chance windows within m substitutions of a query in a proteome of R residues is

$$\mathbb{E}[\#\text{chance}] \approx R \sum_{i=0}^{m} \binom{L}{i} \bar{q}^{\,L-i} (1-\bar{q})^i, \qquad \bar{q} = \textstyle\sum_c q_c^2, \tag{9}$$

the Karlin–Altschul/birthday count specialized to the Hamming ball (the empirical-background variant reuses `seqtree` §KA). For $m \in \{0, 1\}$ and $L \geq 8$, (9) is $\ll 1$, so a hit is essentially certainly the true source rather than coincidence—`Proteome.find_source` returns it with the differing residue flagged.

6. Motif logos and length distributions

For an allele's presented set (class I peptides of a fixed length; class II register-anchored 9-mer cores) the per-position information content is

$$I_i = \log_2 20 - H_i - e(n_i), \qquad H_i = -\sum_c f_{i,c} \log_2 f_{i,c}, \tag{10}$$

with $f_{i,c}$ the residue frequency at position i and $e(n_i)$ the small-sample entropy correction for n_i counts (negligible for the deep `pmhc_data` alleles). The logo draws letter c at height $f_{i,c} I_i$ bits; columns at the anchors stand tall (e.g. A02:01 shows P2=L, PΩ $\in$ {V,L}), the TCR-facing columns are flat. The length distribution is reported as a companion histogram. Columns may be confidence-weighted by n (or by the shortlist $\geq$ 2-publication support). `logo.motif` returns the numeric logo; `logo.render` draws it.

7. Planned predictors and their composition

Presentation is necessary but not sufficient for immunogenicity. Each planned predictor scores a distinct biophysical or population term and *composes* with the presentation evidence on the log scale; for peptide σ and allele a the combined log-odds is

$$\log \frac{\mathbb{P}(\text{immunogenic})}{\mathbb{P}(\text{not})} = \beta_0 + \beta_{\text{pres}} (-\log p_{\text{enrich}}) + \beta_{\text{aff}} \log \tfrac{1}{\text{IC}_{50}} + \beta_{\text{stab}}\, t_{1/2} + \beta_{\text{clv}}\, c_{\text{Cterm}} + \beta_{\text{expr}} \log e_{\text{gene}} + \beta_{\text{imm}}\, \rho_{\text{TCR}}, \tag{11}$$

with coefficients fit on labelled outcome data (the user will supply tuning/benchmark sets). The terms: *affinity* IC_{50} and *stability* $t_{1/2}$ are the quantitative complements of the binary presentation E-value, regressable on the same anchor features plus the pseudosequence; *cleavage* c_{Cterm} is a position-specific C-terminal generation probability (with optional N-terminal trimming); *expression* e_{gene} and variant frequency enter as Bayesian priors multiplying the presentation odds; *immunogenicity* ρ_{TCR} combines physicochemical TCR-facing features with a TCR precursor-frequency estimate in the $Q = R \times D$ spirit of [11]. The precursor-frequency estimator is a TCR-side quantity and is deferred to a separate package, consumed here only as a score. Each term is a roadmap Phase-2 milestone specified by this subsection.

8. Benchmark and evaluation methodology

We evaluate as a per-(peptide, allele) binary task. *Splits* are by allele *and* by peptide: held-out alleles test cross-allele generalization (and the diffusion model of § 4), held-out peptides within an allele test within-allele recall; peptide duplicates are collapsed before splitting, and crucially a held-out peptide is excluded from *every* allele's training set (not just its own), so an identical co-presented copy cannot leak into a score—in particular through cross-allele diffusion. Equivalently, all matching counts are *punctured*: exact-identity hits are dropped, never rewarded (mirroring the $n^{>}$ counts of the `seqtree` appendix). This lowers raw baselines but is the only fair measure of generalization. *Metrics* are ROC-AUC, PR-AUC against the prevalence baseline, top-1 accuracy, and a real-vs-random noise-rejection AUROC (the non-binder filter). *Comparison*: against NetMHCpan-4.1 and NetMHCIIpan-4.0 [13], MixMHCpred [4], and MixMHC2pred [12] on matched alleles and shared eluted-ligand / assay test sets, with rank-threshold calibration and an explicit *rare-allele subgroup* analysis (the equity axis of [8]) where the pseudosequence diffusion is expected to help most. The benchmark datasets and the paper live in a dedicated repository; this section fixes the protocol so it stays consistent with the predictors defined here.

9. Assumptions and limitations

(i) The analytic pooled null assumes approximate position independence in the anchor block; real co-occupancy adds a two-point (b_2) correction (the negative-binomial tail of the `seqtree` appendix) that keeps the mean robust while inflating the tail. (ii) Kernel validity rests on a faithful pseudosequence; ambiguous (`X`) positions are skipped and locus-aware normalization of allele names is required (class II especially). (iii) The pooled E-value is a smoothed estimator (Prop. 2)—its bias must be reported alongside the gain for rare alleles. (iv) The class-II register trick is a one-pass proxy for full deconvolution [3, 2]. (v) "Compatible allele" in mimicry is a user input; cross-reactivity significance is conditional on a faithful per-allele presented control. These are the open items tracked in `ROADMAP.md`.

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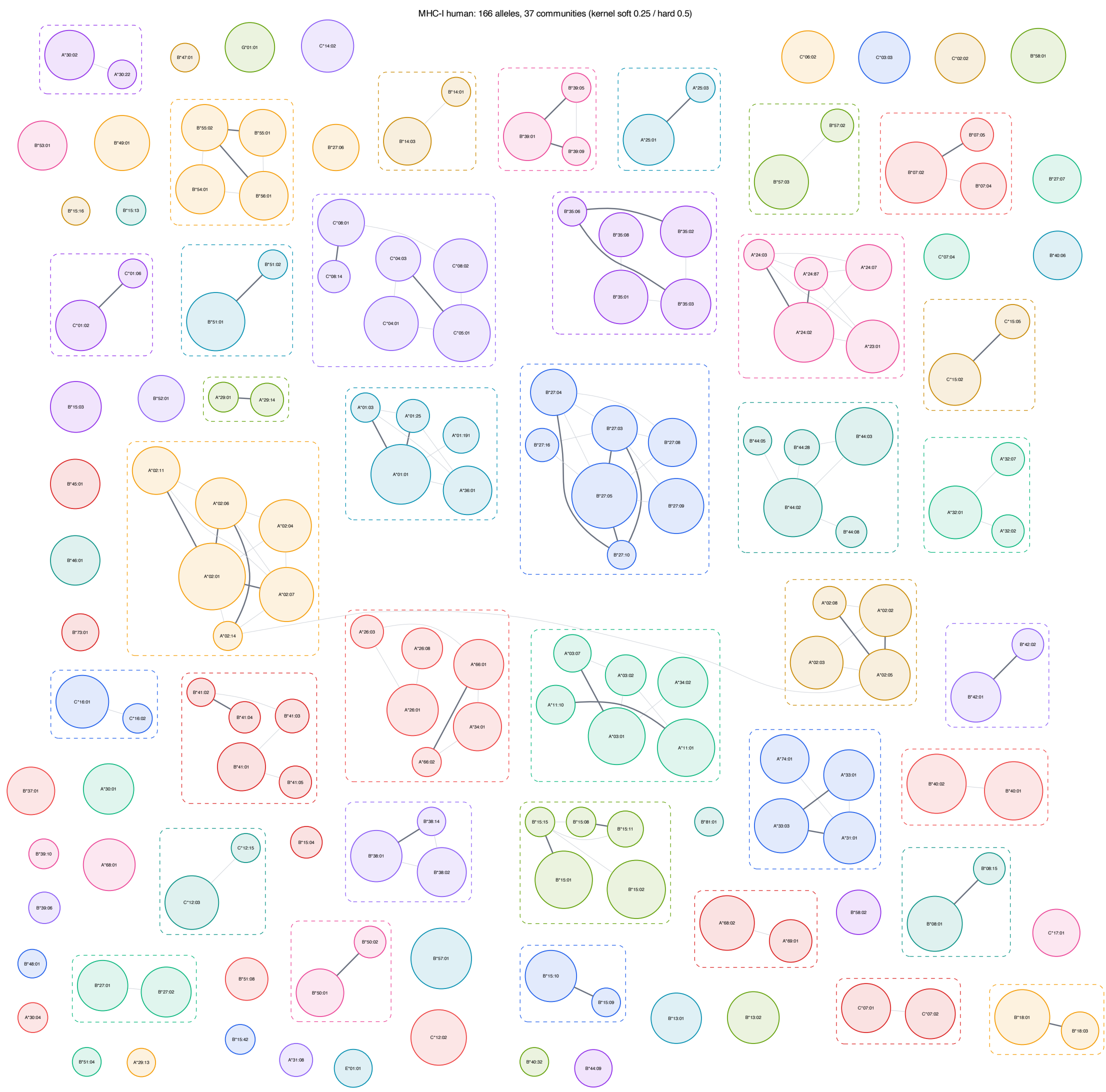

Figure 3: Human MHC-I allele network: nodes are alleles (size $\propto$ presented-set), edges the MI-weighted BLOSUM groove kernel (4) (thin ≥ 0.25, bold ≥ 0.5), dashed boxes the greedy-modularity communities. The communities recover known HLA-I supertypes. Mouse and class-II panels are generated likewise (`bench/promiscuity_graph.py`).

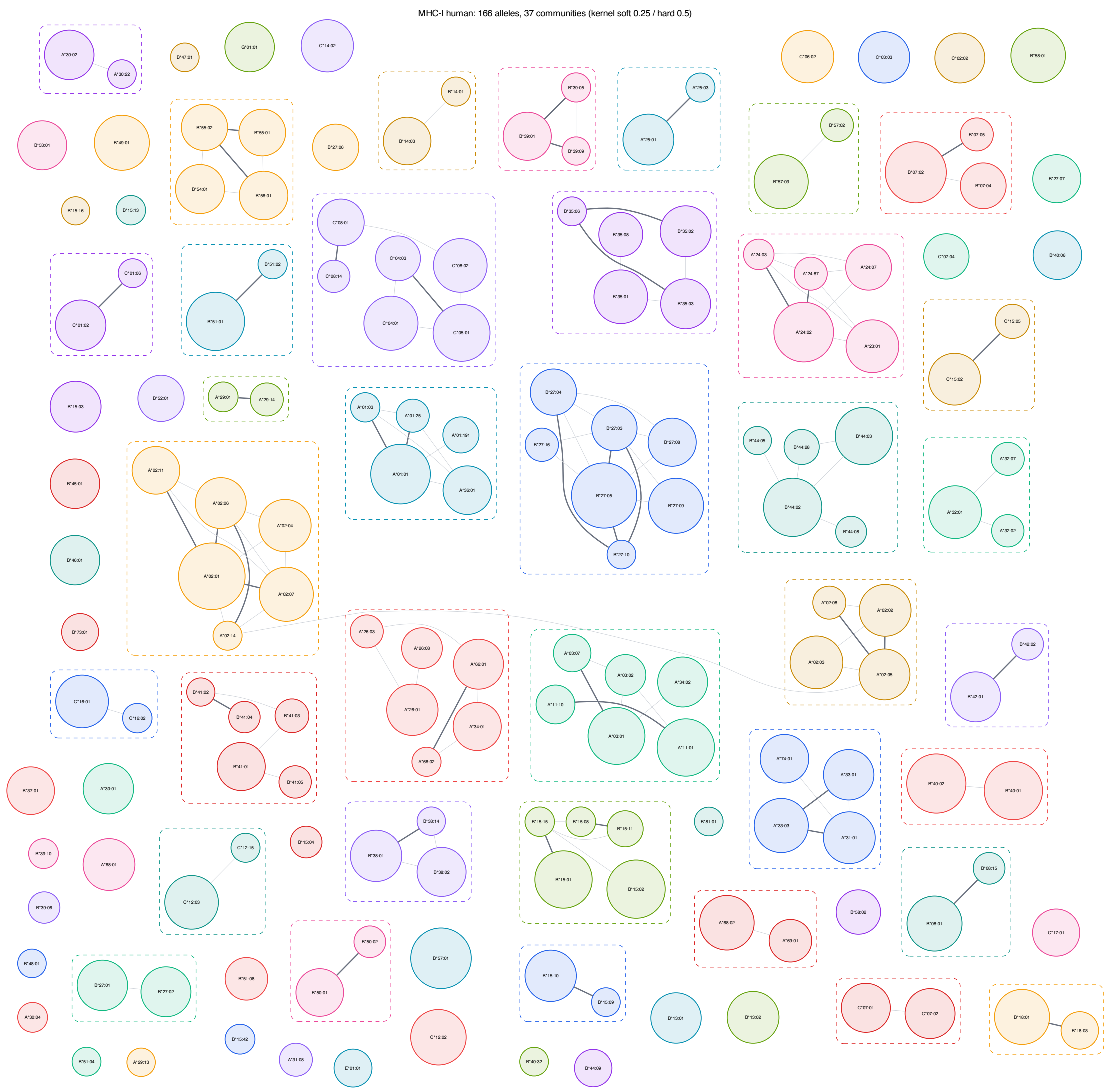

Figure 4: Human MHC-I allele similarity / promiscuity network: nodes are alleles (size ∝ log presented-set size), edges are pseudosequence-kernel similarity (thin above the soft threshold, bold above the hard threshold), dashed boxes are greedy-modularity communities. The communities recover the classical HLA-I supertypes. `bench/promiscuity_graph.py`.

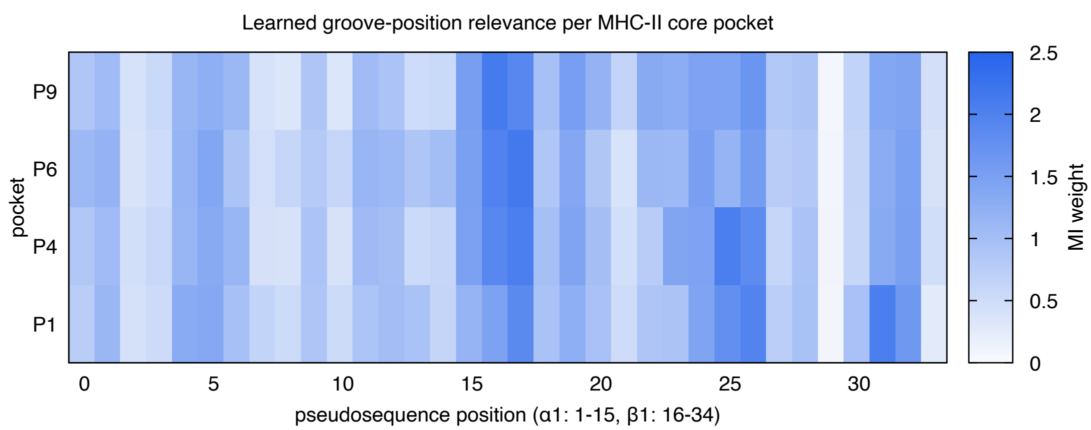

Figure 5: Learned mutual-information relevance $w_{j,p}$ of each groove pseudosequence position p (x) for each MHC-II core pocket j (rows), over human class-II alleles. Relevance concentrates on the polymorphic β_1 region; different pockets emphasize different positions. `bench/make_figures.py`.